

Guidelines

- Kindly use the given template for submitting your prototype (Make a copy of the template)
- One team is only required to submit one project.
- Follow this template while preparing your presentation.
- You are welcome to add as many POCs and design concepts to support your project.
- The project should be feasible and the team members should be capable enough, to come up with the fully functioning prototype of the same idea, if required.

UniHack

AI-Powered Product Intelligence for Industrial Commerce

Team Details

a. Team name: SpecTrace Innovators

b. Team leader name: Rahul Karn

Brief about your solution

SpecTrace turns messy industrial product feeds into search-ready catalog listings:

- Cleans and standardizes raw distributor input, removing placeholder brand tags.
- Sorts products into categories and writes 5 description versions (invoice, mobile, short, long, marketing).
- Fills all 252 required Unilog columns with 100% completion across 1,000 items in CSV and XLSX.
- Disambiguates genuine product brands from supplier/distributor names.

Core Questions

1. How does it enrich minimal product info?

- Extracts dimensions, materials, and series numbers from raw text using structured LLM reasoning.
- Separates real product brands from distributor names and fills up to 50 attribute fields with correct units.

2. How does it ensure accuracy and trust?

- Computes confidence scores per product; flags low-confidence items for automated human review.
- Deterministic check confirms output matches all 252 expected Unilog columns exactly.

3. What makes it scalable for enterprise catalogs?

- Multi-key API pool rotates credentials automatically to stay within rate limits and daily quotas.
- Persistent row-index caching prevents re-processing; modular core allows swapping in other LLMs.

Opportunities

a. How different is it from existing ideas?

- Handles messy real-world data (duplicate part numbers, inconsistent entries) without losing records.
- Matches the exact 252-column enterprise format distributors need, not an approximation.

b. How will it solve the problem statement given?

- Automates the full 8-stage pipeline end to end — from raw feed ingestion to review-ready export.

c. USP of the proposed solution

- Fully automated pipeline hitting the exact enterprise schema with brand separation and zero data loss.

List of features offered by the solution

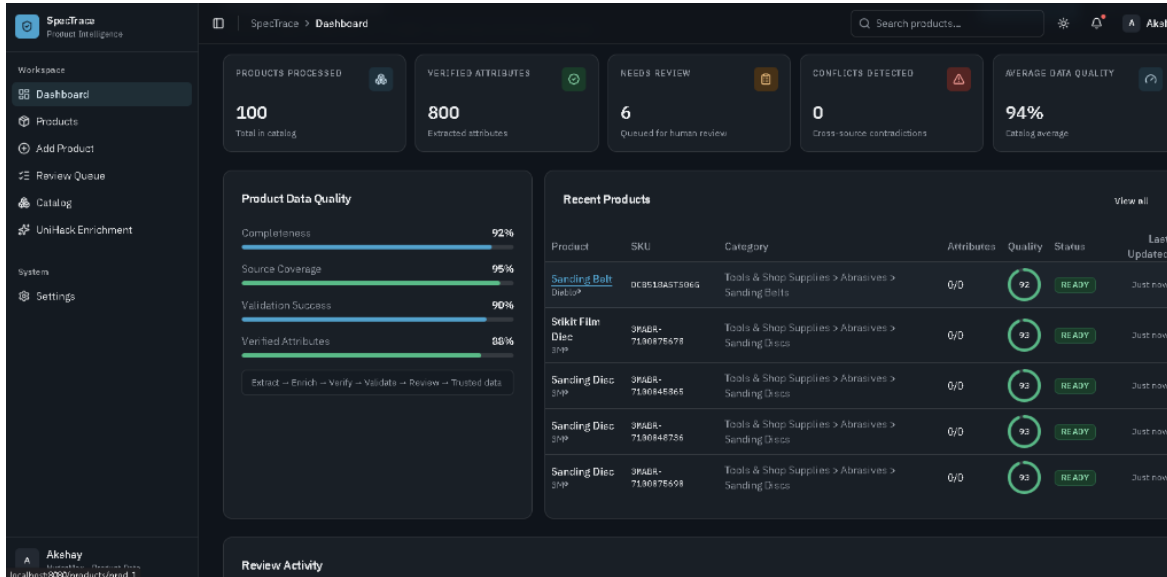
- 1. Automatic 3-Level Product Categorization (Department > Class > Fine taxonomy)**
- 2. Five Description Formats per Product (Invoice <=40 char, Mobile, Short, Long, Marketing)**
- 3. Technical Attribute & Unit (UOM) Normalization Engine**
 4. Real Brand vs. Distributor Name Separation Engine
 5. Multi-Key API Round-Robin Pool with Automatic Quota Rotation
 6. Output Verified Against All 252 Canonical Delivery Columns
 7. Multi-Factor Confidence Scoring with Human Review Flagging
 8. Fast In-Memory Dataset Caching for Instant UI Response
 9. Dual-Format Delivery Exporter (CSV and native multi-sheet XLSX)
 10. Interactive Web Dashboard to Review, Search, and Edit Records

Process flow diagram or Use-case diagram

Raw Feed "" Ingestion & Caching "" Brand Cleanup "" AI Enrichment "" Rule Cleanup & Units "" Confidence Check ""
Final 252-Col Output

- Ingestion & Caching: Raw distributor input loaded and indexed with zero data loss.
- AI Reasoning & Enrichment: Gemini reasoning extracts taxonomy, attributes, and 5 copy tiers.
- Cleansing & Validation: Enforces Invoice ≤ 40 char ALL CAPS and strict unit standards.
- Enterprise Delivery: All 1,000 items formatted into 252 canonical columns.

Wireframes/Mock diagrams of the proposed solution (optional)



- Top Navigation: Real-time catalog KPIs, global product search, and light/dark theme toggle.

- Left Sidebar: Quick navigation between Dashboard, Products, Review Queue, and UniHack Enrichment.

- Main Workspace: Drag-and-drop CSV upload, batch selection table, side-by-side inspector, and 252-col export buttons.

Architecture diagram of the proposed solution

- Client Layer: React 18, TanStack Router, Tailwind CSS, Radix UI

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- API Layer: Node.js Express, TypeScript, Zod Validation, REST Routes

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- Processing Layer: Multi-Key Pool (4 Keys), Gemini 3.5 Flash, JSON Parser, UOM Normalizer

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- Storage & Export: In-Memory Cache, NDJSON Persistence, 252-Col CSV & XLSX Generator

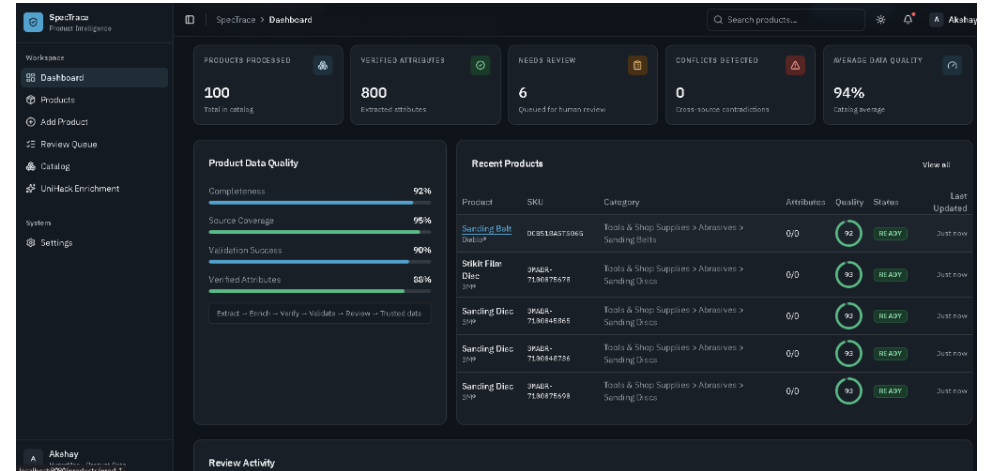
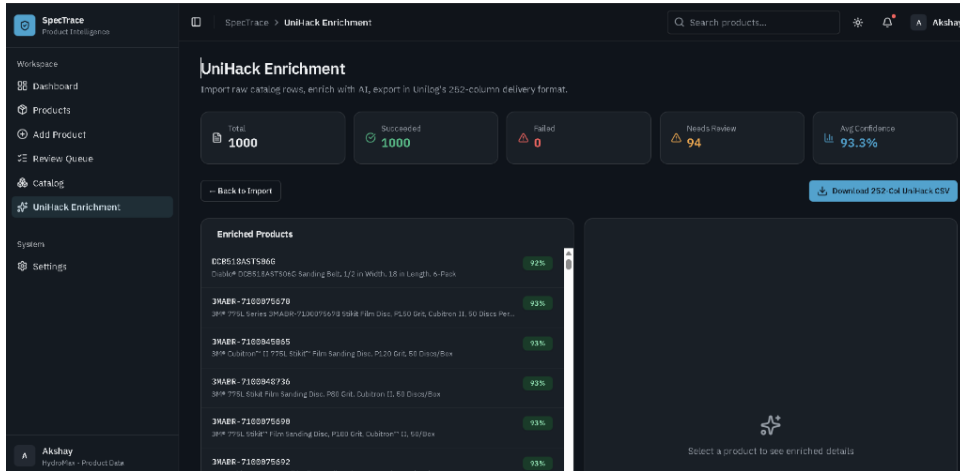
Technologies used in the solution

- Frontend: React 18, Vite, TypeScript, TanStack Router, TanStack Table, Tailwind CSS
- Backend: Node.js, Express, TypeScript, Zod validation, xlsx spreadsheet engine, Pino
- AI Engine: Google Gemini API (@google/genai SDK) with multi-key pool rotation
- Deployment: Render cloud web service, GitHub Actions automated CI/CD pipeline

Estimated implementation cost (optional)

- Cloud Hosting: ~\$7 – \$25 / month (Render Starter / AWS ECS web app hosting)
- AI Inference: ~\$0.075 per 1,000 products using Gemini 3.5 Flash Lite (~\$7.50 per 100k SKUs)
- Cost per Record: ~\$0.0001 per enriched product listing
- Operational Savings: >99.9% cost reduction compared to manual catalog curation teams

Snapshots of the MVP



Batch Enrichment (1,000 Items Loaded, 100% Succeeded, 0 Failed)

Catalog Governance Dashboard (94% Average Quality Score)

Additional Details/Future Development (if any)

1. **Automated OEM Web Crawling:** Auto-crawl manufacturer datasheets for deeper technical specs.
2. **Computer Vision Pipeline:** Automated product image processing and diagram classification.
3. **Direct Enterprise ERP/PIM Connectors:** Turn-key integrations with enterprise catalog platforms (Akeneo, SAP, Shopify).
4. **Custom LOV Taxonomy Validation:** Strict validation against enterprise client master taxonomy and picklist rules.

Provide links to your:

1. GitHub Public Repository

<https://github.com/Akshay-khichi/UniHack>

2. Demo Video Link (3 Minutes)

<https://drive.google.com/drive/folders/12YQEJEIPQPIxFCnwWzJ43MpMh6RsVuiC>

3. Working Prototype Link

<https://unihack-nyab.onrender.com>

UniHack

AI-Powered Product Intelligence for Industrial Commerce

Thank You

Live Demo: <https://unihack-nyab.onrender.com> • GitHub: <https://github.com/Akshay-khichi/UniHack>